higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

27 March 2026 (X-paper)
09:00–12:00

Scientific calculators and drawing instruments may be used.

This question paper consists of 5 pages and a formula sheet of 4 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimal places (unless indicated differently).
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Use only a black or a blue pen.
8. Sketches must be large, neat and fully labelled.
9. Write neatly and legibly.

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QUESTION 1

1.1 Given:

$$\log_3 y = \lim_{x\to 0}\frac{1-\cos^2 x}{x^2}$$

1.1.1 Determine the following limit:

$$\log_3 y = \lim_{x\to 0}\frac{1-\cos^2 x}{x^2}$$ (3)

1.1.2 Find the value of y. (1)

1.2 Determine the interval on which $f(x)$ is continuous if:

$$f(x) = \frac{\sin x}{x+3}$$ (2)

[6]

QUESTION 2

2.1 Given:

$$f(x) = \frac{1-x}{x+2}$$

Determine the derivative of $f(x)$ from first principles. (5)

2.2 Determine $\frac{dy}{dx}$ in each of the following cases:
(Simplification is NOT required)

2.2.1 $y = (x^2+5)^{-\frac{3}{2}}(\sec 5x^2)$ (4)

2.2.2 $y = \frac{1-\sec(\theta^2 x)}{\theta^3 x^3 - \cot\theta}$; where θ is a constant (4)

2.3 With the aid of logarithmic differentiation, calculate $\frac{dy}{dx}$ if $= [\ln(x^2+7)]^{\cos^{-1}x}$. (4)

2.4 Given the implicit function

$$\cos(x^2y^3) + 3x = y\cot 2x,$$

determine $\frac{dy}{dx}$. (5)

[22]

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QUESTION 3

3.1 Given: $f(x) = x^3 + x^2 - x - 5$

3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

3.1.2 Draw up a table of x and $f(x)$, where x is ranging from $x = -2$ to $x = 2;\ \ x \in Z$. (2)

3.1.3 Draw a neat graph of $f(x)$ showing the turning point between these values. (2)

3.1.4 Given that an estimate of the root of the equation $x^3 + x^2 - x - 5 = 0$ is close to $x = 1{,}5$, use Taylor's/Newton's method TWICE to determine a better approximation of this root. (Provide the correct root in THREE decimal figures.) (4)

3.2 A box with a square base and no top is to hold a volume 100 cm^3. Find the dimensions of the box that would require the least material for the five sides.

HINT: $V = l^2h \quad and \quad A = 4lh + l^2$ (5)

3.3 Suppose that the radius of a sphere is changing at an instantaneous rate of 0,1 cm/sec. At what rate is the area of a sphere growing when r = 10 cm?

HINT: $A = 4\pi r^2$ (3)

[19]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $y = \dfrac{x}{\sqrt{1 - 25x^4}}$ (3)

4.1.2 $y = \sin^3 x \cos^4 x$ (5)

4.1.3 $y = \dfrac{2x^3 + x^2 - 4x + 1}{x - 2}$ (5)

4.1.4 $y = xe^{4x}$ (4)

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$$y = \frac{2x^2 - 10}{(x-4)^3}$$ (7)

[24]

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QUESTION 5

5.1 Determine

$$\int_0^{\infty} \frac{1}{\sqrt{3-x}}\,dx$$ (4)

5.2 Given: $y = 10 - x$ and $y = -x^2 + 10x - 8$

5.2.1 Calculate the coordinates of the points of intersection. (3)

5.2.2 Make a neat sketch to show the enclosed area, the representative strip and the points of intersection. (2)

5.2.3 Calculate the magnitude of the area in QUESTION 5.2.2. (4)

5.2.4 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.2.2 is rotated about the x-axis. (5)

5.3 Calculate the second moment of area of a circular lamina with diameter 14 cm about an axis through its centre and perpendicular to the plane of the lamina. (4)

[22]

QUESTION 6

6.1 Determine the particular solution of:

$$x^2 e^y \frac{dy}{dx} = 1\,; \qquad y = 0 \;\; when \;\; x = 1$$ (4)

6.2 Determine the general solution of:

$$e^{-x}\frac{d^2y}{dx^2} = e^{4x} + e^{-x}\sin 3x$$ (3)

[7]

TOTAL: 100

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MATHEMATICS N5

FORMULA SHEET

Any other applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \text{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$\tan 2A = \dfrac{2 \tan A}{1 - \tan^2 A}$

$\sin^2 A = {}^1/_2 - {}^1/_2 \cos 2A$

$\cos^2 A = {}^1/_2 + {}^1/_2 \cos 2A$

$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos(A \pm B) = \cos A \cos B \pm \sin A \sin B$

$\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \pm \tan A \tan B}$

$\sin A \cos B = {}^1/_2 [\sin(A + B) + \sin(A - B)]$

$\cos A \sin B = {}^1/_2 [\sin(A + B) - \sin(A - B)]$

$\cos A \cos B = {}^1/_2 [\cos(A + B) + \cos(A - B)]$

$\sin A \sin B = {}^1/_2 [\cos(A - B) - \cos(A + B)]$

$\tan x = \dfrac{\sin x}{\cos x}$; $\sin x = \dfrac{1}{cosec\ x}$; $\cos x = \dfrac{1}{\sec x}$

BINOMIAL THEOREM

$(x + h)^n = x^n + \text{n}x^{n-1} h + \dfrac{n(n-1)}{2!} x^{n-2} h^2 + \cdots$

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DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x) \,.\, v(x)$$

$$\frac{dy}{dx} = u \,.\, \frac{dv}{dx} + v \,.\, \frac{du}{dx}$$

$$= u \,.\, v' + v \,.\, u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v \,.\, \frac{du}{dx} - u \,.\, \frac{dv}{dx}}{v^2}$$

$$= \frac{v \,.\, u' - u \,.\, v'}{v^2}$$

CHAIN RULE

$$y = f\big(u(x)\big)$$

$$\frac{dy}{dx} = \frac{dy}{du} \,.\, \frac{du}{dx}$$

$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x . lna$	$\frac{a^x}{lna} + c$
$\log_e x$	$\frac{1}{x}$	___
$\log_a x$	$\frac{1}{xlna}$	___
$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$

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$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
$\tan x$	$\sec^2 x$	$ln(\sec x) + c$
$\cot x$	$-\operatorname{cosec}^2 x$	$ln(\sin x) + c$
$\sec x$	$\sec x \,.\tan x$	$ln[\sec x + \tan x] + c$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \,.\cot x$	$ln(\operatorname{cosec} x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1} x$	$\frac{1}{1+x^2}$	—
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	—
$\sec^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$	—
$\operatorname{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	—
$\frac{1}{x\sqrt{x^2+a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2 - x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2 - x^2} \; + c$
$\frac{1}{x^2-a^2}$	—	$\frac{1}{2a} ln\left(\frac{x-a}{x+a}\right) + c$
$\frac{1}{a^2-x^2}$	—	$\frac{1}{2a} ln\left(\frac{a+x}{a-x}\right) + c$

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INTEGRATION

$$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x) g\,(x) dx$$

$$\int \frac{f'(x)}{f(x)} dx = ln\, f(x)\ + c$$

$$\int [f(x)]^n\, f'(x) dx = \frac{[f(x)]^{n+1}}{n\ + 1} + c$$

$$\frac{f(x)}{(ax + b)(cx\ + d)} = \frac{A}{ax\ + b} + \frac{B}{cx\ + d}$$

$$\frac{f(x)}{(x\ + a\ )^n} = \frac{A}{(x\ + a)} + \frac{B}{(x\ + a)^2} + \frac{C}{(x + a)^3} + \cdots \frac{Z}{(x\ + a)^n}$$

APPLICATIONS OF INTEGRATION

AREAS

$$A_x = \int_a^b y dx\ ;\ A_x = \int_a^b (y_1 - y_1) dx$$

$$A_x = \int_a^b x dy\ ; A_{Y=} \int_a^b (x_1 - x_2) dy$$

VOLUMES

$$V_x = \pi \int_a^b y^2\, dx\ ;\ V_{X\,=\,\pi} \int_a^b (y_1^2 - y_2^2) dx$$

$$V_y = \pi \int_a^b x^2\, dy\ ;\ V_{y=\,\pi} \int_a^b (x_1^2 - x_2^2) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2\, dA\ ;\ I_{y=} \int_a^b r^2\, dA$$

MOMENT OF INERTIA

$$Mass\ = density\ \times volume$$

$$M\ = pv$$

$$DEFINITION : I\ = m\, r^2$$

$$GENERAL : I\ = \int_a^b r^2 dm\ = p \int_a^b r^2 dV$$

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